

Analytical Geometry & Polar Coordinates

Exhaustive Step-by-Step Solutions with Mathematical Proofs & Diagrams

Topics: Focal Chords of Conics, Harmonic Mean Property, Polar Lines, and Triangular Areas

Problem 1: Focal Chords of an Ellipse

If PSQ and $PS'R$ be two chords of an ellipse through the foci S and S' respectively, then prove that

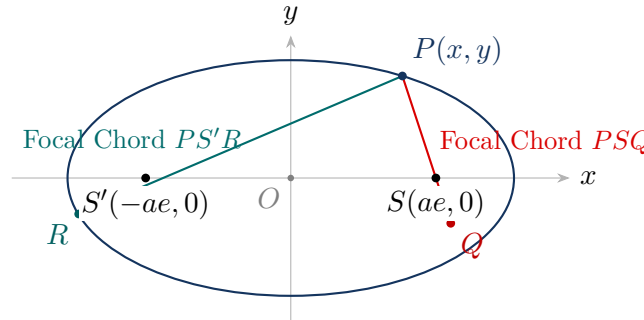
$$\frac{SP}{SQ} + \frac{S'P}{S'R}$$

is independent of the position of P .

1. Geometric Setup and Mathematical Definitions

Consider an ellipse centered at the origin with standard Cartesian equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b > 0$):

- **Parameters:** Semi-major axis a , semi-minor axis b , eccentricity $e = \sqrt{1 - b^2/a^2} < 1$.
- **Semi-latus Rectum:** $l = \frac{b^2}{a} = a(1 - e^2)$.
- **Foci:** $S(ae, 0)$ and $S'(-ae, 0)$ situated symmetrically on the major axis.
- **Focal Chords:** PSQ is a chord through focus S , and $PS'R$ is a chord through focus S' .



2. The Harmonic Mean Property of Focal Chords

Take the focus S as the pole and the major axis as the initial line. The polar equation of the ellipse is:

$$\frac{l}{r} = 1 - e \cos \theta$$

Let the vectorial angle of P with respect to S be α . The radius vector is $SP = r_P$:

$$\frac{l}{SP} = 1 - e \cos \alpha \implies \frac{1}{SP} = \frac{1 - e \cos \alpha}{l}$$

Since PSQ is a straight line through the pole S , point Q has vectorial angle $\alpha + \pi$. Therefore:

$$\frac{l}{SQ} = 1 - e \cos(\alpha + \pi) = 1 + e \cos \alpha \implies \frac{1}{SQ} = \frac{1 + e \cos \alpha}{l}$$

Adding these two reciprocal equations yields the fundamental harmonic mean relation:

$$\frac{1}{SP} + \frac{1}{SQ} = \frac{(1 - e \cos \alpha) + (1 + e \cos \alpha)}{l} = \frac{2}{l} \quad (1)$$

3. Expressing Segment Ratios and Establishing Invariance

Segment Ratio for Chord PSQ Through Focus S : Multiplying both sides of the harmonic mean relation (1) by SP :

$$SP \left(\frac{1}{SP} + \frac{1}{SQ} \right) = SP \cdot \frac{2}{l} \implies 1 + \frac{SP}{SQ} = \frac{2 \cdot SP}{l} \implies \frac{SP}{SQ} = \frac{2 \cdot SP}{l} - 1$$

Segment Ratio for Chord $PS'R$ Through Focus S' : Applying identical reasoning with the second focus S' as the pole (the semi-latus rectum at S' is also $l = b^2/a$):

$$\frac{1}{S'P} + \frac{1}{S'R} = \frac{2}{l} \quad (2)$$

Multiplying both sides of equation (2) by $S'P$:

$$1 + \frac{S'P}{S'R} = \frac{2 \cdot S'P}{l} \implies \frac{S'P}{S'R} = \frac{2 \cdot S'P}{l} - 1$$

Summing the Two Segment Ratios: Adding the two expressions:

$$\frac{SP}{SQ} + \frac{S'P}{S'R} = \left(\frac{2 \cdot SP}{l} - 1 \right) + \left(\frac{2 \cdot S'P}{l} - 1 \right) = \frac{2(SP + S'P)}{l} - 2$$

By the fundamental bifocal property of an ellipse, the sum of the focal distances from any point P on the ellipse to the two foci is constant and identically equal to the length of the major axis $2a$:

$$SP + S'P = 2a$$

Substituting $SP + S'P = 2a$ into the summation formula:

$$\frac{SP}{SQ} + \frac{S'P}{S'R} = \frac{2(2a)}{l} - 2 = \frac{4a}{l} - 2$$

Substituting the value of the semi-latus rectum $l = \frac{b^2}{a} = a(1 - e^2)$:

$$\frac{SP}{SQ} + \frac{S'P}{S'R} = \frac{4a}{\left(\frac{b^2}{a}\right)} - 2 = \frac{4a^2}{b^2} - 2 = \frac{4}{1 - e^2} - 2 = \frac{2(1 + e^2)}{1 - e^2}$$

Conclusion for Problem 1

The evaluated expression:

$$\frac{SP}{SQ} + \frac{S'P}{S'R} = \frac{2(1 + e^2)}{1 - e^2} = \frac{4a^2}{b^2} - 2$$

is a fixed constant determined exclusively by the ellipse parameters (a, b, e) , and is **strictly independent of the position of point P** . ■

Physical and Geometric Significance

Regardless of where P traverses along the ellipse (at the ends of the major axis, ends of the minor axis, or any arbitrary location), as the chord through S lengthens or shortens, the compensating chord through S' adjusts in exact reciprocal proportion such that the combined ratio sum remains invariant.

Problem 2: Angle Between Tangents at Extremities of a Focal Chord

PSP' is a focal chord of the conic. Prove that the angle between the tangents at P and P' is

$$\tan^{-1} \left(\frac{2e \sin \alpha}{1 - e^2} \right)$$

where α is the angle between the chord and the major axis.

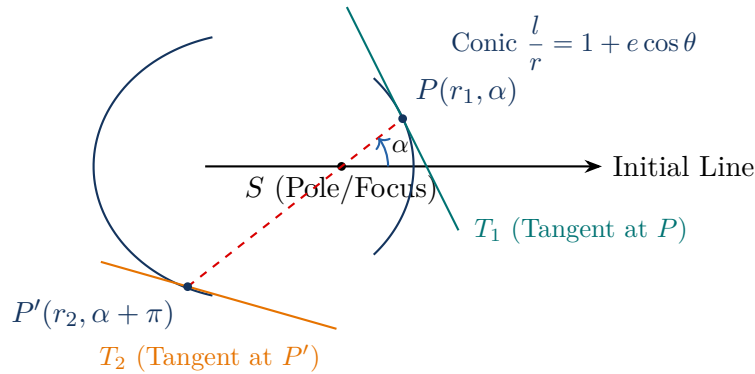
1. Polar Equation of Conic and Extremities of the Chord

Let the focus S be the pole and the major axis (axis of the conic) be the initial line. The polar equation of the conic is:

$$\frac{l}{r} = 1 + e \cos \theta$$

where l is the semi-latus rectum and e is the eccentricity. The focal chord PSP' passes through S and makes an angle α with the initial line:

- Point P has vectorial angle $\theta_P = \alpha$.
- Point P' lies on the opposite ray through the pole, with vectorial angle $\theta_{P'} = \alpha + \pi$.



2. General Polar Tangent Equation and Slopes

The equation of the tangent to the conic $\frac{l}{r} = 1 + e \cos \theta$ at a point with vectorial angle β is:

$$\frac{l}{r} = e \cos \theta + \cos(\theta - \beta) = (e + \cos \beta) \cos \theta + \sin \beta \sin \theta$$

In Cartesian coordinates, where $x = r \cos \theta$ and $y = r \sin \theta$:

$$(e + \cos \beta)x + (\sin \beta)y = l \quad (3)$$

The slope of this tangent line is:

$$m = -\frac{\text{coefficient of } x}{\text{coefficient of } y} = -\frac{e + \cos \beta}{\sin \beta}$$

Slope of Tangent at P ($\beta = \alpha$):

$$m_1 = -\frac{e + \cos \alpha}{\sin \alpha}$$

Slope of Tangent at P' ($\beta = \alpha + \pi$): Using $\cos(\alpha + \pi) = -\cos \alpha$ and $\sin(\alpha + \pi) = -\sin \alpha$:

$$m_2 = -\frac{e + \cos(\alpha + \pi)}{\sin(\alpha + \pi)} = -\frac{e - \cos \alpha}{-\sin \alpha} = \frac{e - \cos \alpha}{\sin \alpha}$$

3. Evaluation of the Angle Between the Two Tangents

Let ϕ be the angle between the two tangents. By the standard slope formula:

$$\tan \phi = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

Step 3.1: Difference of Slopes ($m_2 - m_1$)

$$m_2 - m_1 = \frac{e - \cos \alpha}{\sin \alpha} - \left(-\frac{e + \cos \alpha}{\sin \alpha} \right) = \frac{(e - \cos \alpha) + (e + \cos \alpha)}{\sin \alpha} = \frac{2e}{\sin \alpha}$$

Step 3.2: Product of Slopes and Denominator ($1 + m_1 m_2$)

$$m_1 m_2 = \left(-\frac{e + \cos \alpha}{\sin \alpha} \right) \left(\frac{e - \cos \alpha}{\sin \alpha} \right) = -\frac{e^2 - \cos^2 \alpha}{\sin^2 \alpha}$$

Adding 1 to both sides and using $\sin^2 \alpha + \cos^2 \alpha = 1$:

$$1 + m_1 m_2 = 1 - \frac{e^2 - \cos^2 \alpha}{\sin^2 \alpha} = \frac{\sin^2 \alpha - e^2 + \cos^2 \alpha}{\sin^2 \alpha} = \frac{1 - e^2}{\sin^2 \alpha}$$

Step 3.3: Ratio $\tan \phi$

$$\tan \phi = \frac{m_2 - m_1}{1 + m_1 m_2} = \frac{\frac{2e}{\sin \alpha}}{\frac{1 - e^2}{\sin^2 \alpha}} = \frac{2e}{\sin \alpha} \cdot \frac{\sin^2 \alpha}{1 - e^2} = \frac{2e \sin \alpha}{1 - e^2}$$

Conclusion for Problem 2

Taking the inverse tangent, the angle ϕ between the tangents at P and P' is:

$$\phi = \tan^{-1} \left(\frac{2e \sin \alpha}{1 - e^2} \right)$$



Geometric Remarks & Physical Insights

- Parabola ($e = 1$):** The denominator vanishes ($1 - e^2 = 0$), giving $\tan \phi \rightarrow \infty \implies \phi = 90^\circ$. Thus, the tangents at the extremities of any focal chord of a parabola are always mutually perpendicular and intersect on the directrix.
- Latus Rectum ($\alpha = 90^\circ$):** Since $\sin 90^\circ = 1$, the angle between tangents at the ends of the latus rectum is $\tan^{-1} \left(\frac{2e}{1 - e^2} \right)$.
- Intersection Point Locus:** The point of intersection of the tangents at the ends of a focal chord lies on the corresponding directrix $\frac{l}{r} = e \cos \theta$ for a parabola, and on the directrix line for general conics.

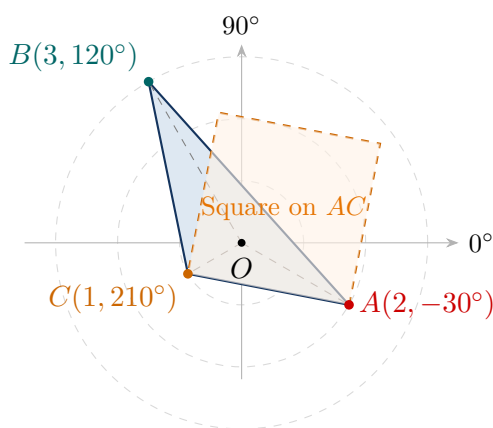
Problem 3: Area of Triangle and Square in Polar Coordinates

Find the area of the triangle whose vertices are $A(2, -30^\circ)$, $B(3, 120^\circ)$, and $C(1, 210^\circ)$. Also find the area of the square whose one side is AC .

1. Given Parameters and Polar Coordinate Diagram

The vertices are given in polar coordinates (r, θ) as:

$$A = (r_1, \theta_1) = (2, -30^\circ), \quad B = (r_2, \theta_2) = (3, 120^\circ), \quad C = (r_3, \theta_3) = (1, 210^\circ)$$



2. Part 1: Area of Triangle ABC (Direct Polar Formula)

The area Δ of a triangle with vertices (r_1, θ_1) , (r_2, θ_2) , and (r_3, θ_3) in polar coordinates is:

$$\Delta = \frac{1}{2} |r_1 r_2 \sin(\theta_2 - \theta_1) + r_2 r_3 \sin(\theta_3 - \theta_2) + r_3 r_1 \sin(\theta_1 - \theta_3)|$$

Computing each term systematically:

$$1. \text{ Term 1: } \theta_2 - \theta_1 = 120^\circ - (-30^\circ) = 150^\circ \implies r_1 r_2 \sin(150^\circ) = (2)(3) \left(\frac{1}{2}\right) = 3$$

$$2. \text{ Term 2: } \theta_3 - \theta_2 = 210^\circ - 120^\circ = 90^\circ \implies r_2 r_3 \sin(90^\circ) = (3)(1)(1) = 3$$

$$3. \text{ Term 3: } \theta_1 - \theta_3 = -30^\circ - 210^\circ = -240^\circ \implies \sin(-240^\circ) = \sin(60^\circ) = \frac{\sqrt{3}}{2} \implies r_3 r_1 \sin(-240^\circ) = (1)(2) \left(\frac{\sqrt{3}}{2}\right) = \sqrt{3}$$

Summing the three terms:

$$\Sigma = 3 + 3 + \sqrt{3} = 6 + \sqrt{3} \implies \text{Area}(\triangle ABC) = \frac{6 + \sqrt{3}}{2} \approx 3.8660 \text{ sq. units}$$

Cartesian Coordinate Verification for Triangle Area: Convert the vertices to Cartesian coordinates using $x = r \cos \theta$ and $y = r \sin \theta$:

- $A(2, -30^\circ) \Rightarrow x_1 = 2 \cos(-30^\circ) = \sqrt{3}, \quad y_1 = 2 \sin(-30^\circ) = -1$
- $B(3, 120^\circ) \Rightarrow x_2 = 3 \cos(120^\circ) = -\frac{3}{2}, \quad y_2 = 3 \sin(120^\circ) = \frac{3\sqrt{3}}{2}$
- $C(1, 210^\circ) \Rightarrow x_3 = 1 \cos(210^\circ) = -\frac{\sqrt{3}}{2}, \quad y_3 = 1 \sin(210^\circ) = -\frac{1}{2}$

Using the standard Shoelace determinant formula:

$$\Delta = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$\begin{aligned} x_1(y_2 - y_3) &= \sqrt{3} \left(\frac{3\sqrt{3} + 1}{2} \right) = \frac{9 + \sqrt{3}}{2}, & x_2(y_3 - y_1) &= -\frac{3}{2} \left(\frac{1}{2} \right) = -\frac{3}{4} \\ x_3(y_1 - y_2) &= -\frac{\sqrt{3}}{2} \left(-1 - \frac{3\sqrt{3}}{2} \right) = \frac{2\sqrt{3} + 9}{4} \end{aligned}$$

Summing the terms inside the absolute value:

$$\frac{18 + 2\sqrt{3} - 3 + 2\sqrt{3} + 9}{4} = \frac{24 + 4\sqrt{3}}{4} = 6 + \sqrt{3} \Rightarrow \Delta = \frac{6 + \sqrt{3}}{2} \quad \checkmark$$

3. Part 2: Area of the Square with Side AC

The square of the side length AC is calculated via the Law of Cosines in polar coordinates:

$$AC^2 = r_1^2 + r_3^2 - 2r_1r_3 \cos(\theta_3 - \theta_1)$$

Here $\theta_3 - \theta_1 = 210^\circ - (-30^\circ) = 240^\circ$. Since $\cos(240^\circ) = -\cos(60^\circ) = -\frac{1}{2}$:

$$AC^2 = 2^2 + 1^2 - 2(2)(1) \left(-\frac{1}{2} \right) = 4 + 1 + 2 = 7$$

Cartesian Distance Check:

$$AC^2 = (x_3 - x_1)^2 + (y_3 - y_1)^2 = \left(-\frac{3\sqrt{3}}{2} \right)^2 + \left(\frac{1}{2} \right)^2 = \frac{27}{4} + \frac{1}{4} = \frac{28}{4} = 7 \quad \checkmark$$

Thus, the area of a square constructed on side AC is:

$$\text{Area of Square} = AC^2 = 7 \text{ sq. units}$$

Final Result for Problem 3

$$\text{Area of Triangle } ABC = \frac{6 + \sqrt{3}}{2} \text{ sq. units}$$

$$\text{Area of Square on side } AC = 7 \text{ sq. units}$$

Combined notation as recorded in original handwritten notes: $\left(\frac{6 + \sqrt{3}}{2}, 7 \right)$.

Problem 4: Polar Equation of a Straight Line Through Two Points

Find the polar equation of the straight line joining the two points $\left(1, \frac{\pi}{2}\right)$ and $(2, \pi)$.

1. Method 1: Using the Two-Point Polar Form

The general equation of a straight line passing through points $P_1(r_1, \theta_1)$ and $P_2(r_2, \theta_2)$ is:

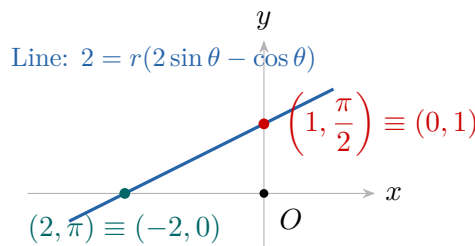
$$\frac{1}{r} \sin(\theta_2 - \theta_1) = \frac{1}{r_1} \sin(\theta_2 - \theta) + \frac{1}{r_2} \sin(\theta - \theta_1) \quad (4)$$

The given points are $(r_1, \theta_1) = \left(1, \frac{\pi}{2}\right)$ and $(r_2, \theta_2) = (2, \pi)$:

- $\theta_2 - \theta_1 = \pi - \frac{\pi}{2} = \frac{\pi}{2} \implies \sin(\theta_2 - \theta_1) = \sin\left(\frac{\pi}{2}\right) = 1$
- $\frac{1}{r_1} \sin(\theta_2 - \theta) = \frac{1}{1} \sin(\pi - \theta) = \sin \theta$
- $\frac{1}{r_2} \sin(\theta - \theta_1) = \frac{1}{2} \sin\left(\theta - \frac{\pi}{2}\right) = -\frac{1}{2} \cos \theta$

Substituting these into formula (4):

$$\frac{1}{r}(1) = \sin \theta - \frac{1}{2} \cos \theta \implies \frac{2}{r} = 2 \sin \theta - \cos \theta \iff 2 = r(2 \sin \theta - \cos \theta)$$

**2. Method 2: Verification via Cartesian Intercept Form**

Convert the given polar points to Cartesian coordinates ($x = r \cos \theta$, $y = r \sin \theta$):

- For $\left(1, \frac{\pi}{2}\right)$: $x_1 = 1 \cos\left(\frac{\pi}{2}\right) = 0$, $y_1 = 1 \sin\left(\frac{\pi}{2}\right) = 1 \implies (0, 1)$ (y -intercept $b = 1$)
- For $(2, \pi)$: $x_2 = 2 \cos \pi = -2$, $y_2 = 2 \sin \pi = 0 \implies (-2, 0)$ (x -intercept $a = -2$)

Using the two-intercept form of a straight line $\frac{x}{a} + \frac{y}{b} = 1$:

$$\frac{x}{-2} + \frac{y}{1} = 1 \implies -x + 2y = 2 \implies 2y - x = 2$$

Substituting the polar relations $x = r \cos \theta$ and $y = r \sin \theta$:

$$2(r \sin \theta) - (r \cos \theta) = 2 \implies r(2 \sin \theta - \cos \theta) = 2$$

Final Answer for Problem 4

The required polar equation of the straight line is:

$$\mathbf{2 = r(2 \sin \theta - \cos \theta)} \quad \text{or} \quad \mathbf{\frac{2}{r} = 2 \sin \theta - \cos \theta}$$

Problem 5: Derivation of the General Polar Equation of a Straight Line

Show that the polar equation of the straight line passing through the points (r_1, θ_1) and (r_2, θ_2) is

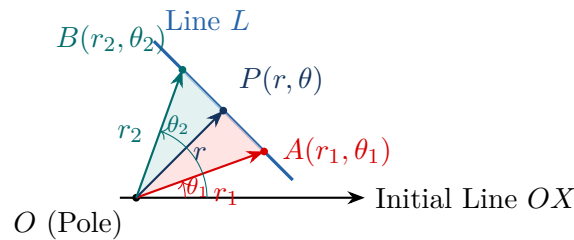
$$\frac{1}{r} \sin(\theta_2 - \theta_1) = \frac{1}{r_1} \sin(\theta_2 - \theta) + \frac{1}{r_2} \sin(\theta - \theta_1)$$

or in symmetric cyclic form:

$$\frac{\sin(\theta_2 - \theta_1)}{r} + \frac{\sin(\theta - \theta_2)}{r_1} + \frac{\sin(\theta_1 - \theta)}{r_2} = 0$$

1. Geometric Configuration

Let O be the pole (origin) and OX the initial line. Let $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$ be two given points on the straight line, with $\theta_2 > \theta_1$. Let $P(r, \theta)$ be an arbitrary point on the straight line situated between A and B , so that $\theta_1 \leq \theta \leq \theta_2$.

**2. Proof Method 1: By Addition of Triangular Areas (Geometric Proof)**

Since the points A , P , and B are collinear, the segment AB is partitioned into AP and PB . Consequently, the area of $\triangle OAB$ is the algebraic sum of the areas of the two sub-triangles $\triangle OAP$ and $\triangle OPB$:

$$\text{Area}(\triangle OAB) = \text{Area}(\triangle OAP) + \text{Area}(\triangle OPB) \quad (5)$$

Recall that the area of any triangle with two sides p, q and included angle ψ is given by $\frac{1}{2}pq \sin \psi$:

1. **In $\triangle OAB$:** Sides $OA = r_1, OB = r_2$, included angle $\angle AOB = \theta_2 - \theta_1$:

$$\text{Area}(\triangle OAB) = \frac{1}{2}r_1r_2 \sin(\theta_2 - \theta_1)$$

2. **In $\triangle OAP$:** Sides $OA = r_1, OP = r$, included angle $\angle AOP = \theta - \theta_1$:

$$\text{Area}(\triangle OAP) = \frac{1}{2}r_1r \sin(\theta - \theta_1)$$

3. **In $\triangle OPB$:** Sides $OP = r, OB = r_2$, included angle $\angle POB = \theta_2 - \theta$:

$$\text{Area}(\triangle OPB) = \frac{1}{2}rr_2 \sin(\theta_2 - \theta)$$

Substituting these three expressions into equation (5):

$$\frac{1}{2}r_1r_2 \sin(\theta_2 - \theta_1) = \frac{1}{2}r_1r \sin(\theta - \theta_1) + \frac{1}{2}rr_2 \sin(\theta_2 - \theta)$$

Divide both sides of the equation by $\frac{1}{2}rr_1r_2$:

$$\frac{r_1r_2 \sin(\theta_2 - \theta_1)}{rr_1r_2} = \frac{rr_2 \sin(\theta_2 - \theta)}{rr_1r_2} + \frac{r_1r \sin(\theta - \theta_1)}{rr_1r_2}$$

Simplifying each fraction:

$$\frac{1}{r} \sin(\theta_2 - \theta_1) = \frac{1}{r_1} \sin(\theta_2 - \theta) + \frac{1}{r_2} \sin(\theta - \theta_1) \quad (6)$$

This completes the direct geometric proof. ■

3. Proof Method 2: Via the General Linear Form and Determinants

The general equation of a straight line in polar coordinates is:

$$\frac{1}{r} = A \cos \theta + B \sin \theta$$

where A and B are constants. Since points $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$ lie on this line:

$$A \cos \theta + B \sin \theta - \frac{1}{r} = 0 \quad (7)$$

$$A \cos \theta_1 + B \sin \theta_1 - \frac{1}{r_1} = 0 \quad (8)$$

$$A \cos \theta_2 + B \sin \theta_2 - \frac{1}{r_2} = 0 \quad (9)$$

For non-trivial constants $(A, B, -1)$, the determinant of coefficients must vanish:

$$\begin{vmatrix} \frac{1}{r} & \cos \theta & \sin \theta \\ \frac{1}{r_1} & \cos \theta_1 & \sin \theta_1 \\ \frac{1}{r_2} & \cos \theta_2 & \sin \theta_2 \end{vmatrix} = 0$$

Expanding along the first column:

$$\frac{1}{r}(\cos \theta_1 \sin \theta_2 - \sin \theta_1 \cos \theta_2) - \frac{1}{r_1}(\cos \theta \sin \theta_2 - \sin \theta \cos \theta_2) + \frac{1}{r_2}(\cos \theta \sin \theta_1 - \sin \theta \cos \theta_1) = 0$$

Applying the sine difference formula $\sin(u - v) = \sin u \cos v - \cos u \sin v$:

$$\frac{1}{r} \sin(\theta_2 - \theta_1) - \frac{1}{r_1} \sin(\theta_2 - \theta) + \frac{1}{r_2} \sin(\theta_1 - \theta) = 0$$

Using $\sin(\theta_1 - \theta) = -\sin(\theta - \theta_1)$:

$$\frac{1}{r} \sin(\theta_2 - \theta_1) = \frac{1}{r_1} \sin(\theta_2 - \theta) + \frac{1}{r_2} \sin(\theta - \theta_1)$$

Rearranging to the cyclic symmetric form by using $\sin(\theta_2 - \theta) = -\sin(\theta - \theta_2)$:

$$\frac{\sin(\theta_2 - \theta_1)}{r} + \frac{\sin(\theta - \theta_2)}{r_1} + \frac{\sin(\theta_1 - \theta)}{r_2} = 0$$

Summary of Equivalent Forms for Problem 5

1. Standard Two-Point Polar Form:

$$\frac{1}{r} \sin(\theta_2 - \theta_1) = \frac{1}{r_1} \sin(\theta_2 - \theta) + \frac{1}{r_2} \sin(\theta - \theta_1)$$

2. Cyclic Symmetric / Determinant Form:

$$\frac{\sin(\theta_2 - \theta_1)}{r} + \frac{\sin(\theta - \theta_2)}{r_1} + \frac{\sin(\theta_1 - \theta)}{r_2} = 0$$

Direct Deduction of Problem 4 from General Theorem 5

Substituting $(r_1, \theta_1) = \left(1, \frac{\pi}{2}\right)$ and $(r_2, \theta_2) = (2, \pi)$ into Theorem 5:

$$\frac{1}{r} \sin\left(\pi - \frac{\pi}{2}\right) = \frac{1}{1} \sin(\pi - \theta) + \frac{1}{2} \sin\left(\theta - \frac{\pi}{2}\right) \implies \frac{1}{r}(1) = \sin \theta - \frac{1}{2} \cos \theta \implies 2 = r(2 \sin \theta - \cos \theta)$$

which precisely recovers the exact solution to Problem 4.